LOAN APPROVAL PREDICTION SYSTEM

Project Problem Statement & Data Dictionary

1. Project Background

In the modern banking and financial sector, managing credit risk effectively is one of the most critical challenges. A mid-sized financial institution offering personal and home loans to customers across urban and rural regions of India receives hundreds of applications daily through online and branch channels. Historically, the institution relied on manual verification, where loan officers manually evaluated applications by checking income proofs, employment details, credit history, and other documents. This process is slow, prone to human bias, and inconsistent across different branches.

2. Business Problem & Challenges

Due to the inefficiencies of manual underwriting, the financial institution faces two major operational challenges:

- **Loss of Business:** Eligible and creditworthy ("good") customers are sometimes mistakenly rejected, driving them to competitors.
- **Credit Risk & Financial Loss:** High-risk applicants are occasionally approved, leading to loan defaults and a rise in Non-Performing Assets (NPAs).

To resolve this, the institution aims to transition to an intelligent, automated loan approval system powered by Machine Learning. The system will automatically analyze applicant profiles and predict whether a loan should be Approved or Rejected before final human verification, ensuring fast, objective, and data-driven decisions.

Machine Learning Target: This is structured as a **Binary Classification** task:

- **Approved (1):** Low-risk profiles likely to repay on time.
- **Rejected (0):** High-risk profiles with a strong likelihood of default.

3. Dataset Description

The historical dataset consists of previous customer loan applications. Each row represents a unique applicant profile containing personal, financial, and credit attributes:

Column Name	Description & Permissible Values
Applicant_ID	Unique identification token for each applicant
Applicant_Income	Monthly income of the primary applicant
Coapplicant_Income	Monthly income of the co-applicant

Column Name	Description & Permissible Values
Employment_Status	Employment type (Salaried / Self-Employed / Business)
Age	Age of the applicant in years
Marital_Status	Marital status (Married / Single)
Dependents	Number of financial dependents
Credit_Score	Credit bureau / Bureau historical score
Existing_Loans	Number of active ongoing loans
DTI_Ratio	Debt-to-Income ratio
Savings	Current savings account balance
Collateral_Value	Estimated valuation of the submitted collateral asset
Loan_Amount	Total loan financial amount requested
Loan_Term	Requested loan tenure/duration in months
Loan_Purpose	Category of loan (Home / Education / Personal / Business)
Property_Area	Geographical location type (Urban / Semi-Urban / Rural)
Education_Level	Highest educational tier (Graduate / Postgraduate / Undergraduate)
Gender	Gender of the applicant (Male / Female)
Employer_Category	Sector of the employer organization (Govt / Private / Self)
Loan_Approved (Target)	Status output (1 = Approved, 0 = Rejected)

4. Technical Implementation Scope

- **Exploratory Data Analysis (EDA):** Conduct data cleaning, profile missing parameters, handle outliers, and inspect cross-correlations.
- **Feature Engineering:** Encode categorical metrics, implement feature scaling/normalization, and handle potential class imbalances.
- **Supervised ML Pipeline:** Develop, train, and test binary classifiers using K-Nearest Neighbors (KNN), Logistic Regression, and Naive Bayes models.
- **Model Evaluation:** Assess classifier reliability primarily through **Precision**, **Recall**, and **F1-Score** to optimize credit default risk controls.